

Portable Aerosol Spectrometer MODEL 11-D



Grimm contact information:

Grimm Aerosol Technik Ainring GmbH & Co.KG
Dorfstraße 9
83404 Ainring
Germany

website: www.grimm-aerosol.com

e-mail: info@grimm-aerosol.com

phone: +49 8654-578-0

fax.:+49 8654-578-35

Trademark and copyright information:

Many software and hardware names in this manual are registered trademarks and must be considered as such.

© 2020 Grimm Aerosol Technik Ainring GmbH & Co.KG. All rights reserved.

No part of this manual may be reproduced in any form or by any means without the written permission of Grimm Aerosol Technik Ainring GmbH & Co. KG (print, copy microfilm, other media) or duplicated, reproduced or modified using electronic systems.

The current **Manual_E_11-D_V2-2_(07-08-20).docx** refers to the model **Model 11-D**.

Contents

1	Introduction	6
2	Basic Operating Requirements	7
2.1	Explanation of symbols and special text characters	7
2.2	General.....	7
2.3	Warranty.....	8
2.4	Safety requirements	9
2.5	Electrical safety	9
2.6	Laser safety.....	9
2.7	Applications and appropriate use.....	10
2.8	Application restrictions	11
2.9	Transport.....	11
2.10	Storing.....	11
2.11	Disposal	12
2.12	Repair.....	12
3	Technical data.....	13
3.1	Measuring principle	13
3.2	Technical data.....	14
3.3	Scope of delivery and optional accessories.....	15
4	Instrument description.....	17
4.1	Operating controls.....	17
4.2	Display.....	18
4.3	On / off switch, standby and measurement mode	18
4.4	Keyboard.....	19
4.5	Sample air inlet	22
4.6	Analog input	22
4.7	Socket for power supply unit.....	23
4.8	Status LEDs	23
4.9	Filter compartment with bayonet lock	23
4.10	Tripod socket.....	23
4.11	Sample air outlet	23
4.12	Battery compartment.....	23
4.13	GPS connection	24
4.14	Bluetooth antenna	24
4.15	Name plate, calibration sticker and security label.....	24
4.16	Data storage / memory with USB flash drive	24
4.17	Data interfaces	25
5	Start-up and measurement procedure	26
5.1	Measured values on the display	26
6	Online data acquisition.....	28
6.1	Spectrometer Software 1179	28
6.2	Modbus.....	30
6.3	HyperTerminal.....	32
7	Maintenance and cleaning	35
7.1	Maintenance and cleaning by the customer	35

8	Appendix	36
8.1	Standard protocol.....	36
8.2	RS-232 Commands.....	40
8.3	Calculation of the dust mass fractions	45
9	Index.....	47

Figures and Illustrations

<i>Figure 2-1: Laser Safety - Warnings</i>	<i>10</i>
<i>Figure 3-1: Schematic structure of the optical measuring cell</i>	<i>13</i>
<i>Figure 3-2: Scope of delivery 11-D (set) accessories</i>	<i>15</i>
<i>Figure 3-3: Optional accessories.....</i>	<i>16</i>
<i>Figure 4-1: Front view.....</i>	<i>17</i>
<i>Figure 4-2: Rear view</i>	<i>17</i>
<i>Figure 4-3: Side view left.....</i>	<i>17</i>
<i>Figure 4-4: Side view right.....</i>	<i>17</i>
<i>Figure 4-5: Keyboard.....</i>	<i>19</i>
<i>Figure 4-6: Pin assignment of the "RS-232" socket on the front of the aerosol spectrometer.....</i>	<i>25</i>
<i>Figure 6-1: Start screen.....</i>	<i>28</i>
<i>Figure 6-2: Selection of the interface</i>	<i>28</i>
<i>Figure 6-3: Selection and initialization of the storage interval in the Time Interval menu.</i>	<i>29</i>
<i>Figure 6-4: User Settings Menu</i>	<i>29</i>
<i>Figure 6-5: In the control window all menu items are active and highlighted in light gray</i>	<i>29</i>
<i>Figure 6-6: Initialization of the measurement by pressing "Start Device"</i>	<i>29</i>
<i>Figure 6-7: New HyperTerminal connection.....</i>	<i>32</i>
<i>Figure 6-8: Interface selection.....</i>	<i>33</i>
<i>Figure 6-9: Transmission parameter</i>	<i>33</i>

Tables

<i>Table 1: Technical data of 11-D</i>	<i>14</i>
<i>Table 2: Scope of delivery 11-D (set) accessories.....</i>	<i>15</i>
<i>Table 3: Optional accessories</i>	<i>16</i>
<i>Table 4: Controls</i>	<i>18</i>
<i>Table 5: Key functions before response to filter change</i>	<i>19</i>
<i>Table 6: Key functions in standby</i>	<i>20</i>
<i>Table 7: Key functions in measuring mode</i>	<i>21</i>
<i>Table 8: Warning and error messages on the display.....</i>	<i>27</i>
<i>Table 9: Overview of the supported function codes.....</i>	<i>30</i>
<i>Table 10: Assignment of holding registers in Modbus</i>	<i>31</i>
<i>Table 11: Coding of the interval</i>	<i>31</i>
<i>Table 12: Coding of the operating status</i>	<i>32</i>
<i>Table 13: Parameters for calculation of the dust mass fraction in the 11-D</i>	<i>45</i>

Abbreviations

CEN	European Committee for Standardization (French Comité Européen de Normalisation)
DIN	German Institute for Standardization. DIN standards can be national (DIN), European (DIN EN) or international (DIN EN ISO).
EDP	Electronic data processing
EMC	Electromagnetic compatibility. Interference by electric or electromagnetic effects on or by other devices
EN	European standard. A rule resulting from a public standardization process and enacted by a European Committee for Standardization (e. g. CEN).
ISO	International Organization for Standardization. Association of standardization organizations that develop international standards
LAN	A Local Area Network (LAN) is a computer network in which various devices (computers, printers, fax machines, etc.) are interconnected.
MODBUS	Communication protocol with master / slave architecture (master = PC, slave = measuring system)
OLED	Organic-Light-Emitting-Diode (Display)
PM	Particulate Matter, dust
pm	Particulate Matter, dust. pm_{10} , $pm_{2.5}$ and pm_1 dustmass fractions are calculated with a different algorithm than PM_{10} , $PM_{2.5}$ and PM_1
PTFE	Polytetrafluoroethylene, a plastic made of fluorine and carbon, also known as Teflon.
REACH	Registration, Evaluation, Authorization and Restriction of Chemicals, the European Union Regulation on the protection of human health and the environment against the risks posed by chemicals
RoHS	Restriction of Hazardous Substances
SMPS+C	Scanning Mobility Particle Sizer with condensation particle counter
SVC	Semi-Volatile-Compounds, usually hydrocarbons that are gaseous at ambient temperature or condense on particle surfaces. Depending on the aerosol, SVC can contribute significantly to the PM and can be lost during sampling, depending on the device type and design.
TC	Total Counts, total particle concentration
TCP/IP	Transmission Control Protocol/Internet Protocol, network protocol
TSP	Total Suspended Particles, dust mass fraction without standard reference. Considers all particles < approx. 50 to 100 μm
USB	Universal serial bus, serial connection for data transfer between a computer and external devices
VDE	Association of Electrical Engineering Electronics Information Technology e.V., Network Organization

1 Introduction

The 11-D is a portable aerosol spectrometer to determine the particle size distribution and the dust mass fractions. The instrument has an integrated gravimetric filter collecting all particles after optical detection. These particles are available for further chemical, microscopical or other investigations. The instrument can be operated with a battery for mobile use, or for stationary use by means of a power supply.

The 11-D is the newest development in the successful model series of the portable aerosol spectrometers by GRIMM. It combines advantages of the previous devices with substantial improvements of the hard- and software.

Benefits of the 11-D:

- improved optical cell, 31 size channels, equidistant
- calibration according to ISO 21501-1
- improved electronic
- completely revised battery for more than 10 hours operation
- charging of the battery with separate table charger in 3 hours only
- optional GPS-signal for position tracking during mobile measurements
- output of twelve dust mass fractions TSP, PM₁₀, PM₄, PM_{2.5}, PM₁, PM_{Coarse}, inhalable, thoracic, respirable, pm₁₀, pm_{2.5}, pm₁ as well as total number concentration and size distribution
- measurement of temperature and relative humidity inside the optical cell
- automated sample flow correction by means of ambient pressure measurement
- flexible interfaces (RS-232, USB (type B), Ethernet (TCP-IP), Bluetooth and USB-storage
- Modbus

In addition, the established advantages of the previous model series were retained, e.g.:

- 1-second interval
- GRIMM Dual-Technology with integrated PTFE filter for gravimetric validation of the calculated dust mass
- internal rinsing air circuit with particle free air to protect the optical components
- self-test of all optical and pneumatic components at start of the measurement
- compact and robust design

2 Basic Operating Requirements

2.1 Explanation of symbols and special text characters

The following graphical symbols facilitate your search for important information:



This symbol indicates useful tips for easy handling of the instrument.



This symbol indicates potential hazards that can cause malfunction or even damage to the hardware and the methods to avoid them.

The entries via instrument keyboard or the entries via PC keyboard are indicated in bold letters in curly brackets.

- e. g. by simultaneously pressing the **{VALUES / DISPLAY}** key with the **{+}** key on the unit ...
by pressing the key sequence **{ | }** and **{Tab}** on the PC over HyperTerminal ...

2.2 General

Many software and hardware names in this manual are registered trademarks and as such must be considered. The author has attempted to provide accurate and complete information. Grimm Aerosol Technik gives no guarantee for the completeness and accuracy of the information provided, and shall not be liable for any damages caused directly or indirectly by the use of this information and/or the use of hardware and software. Grimm Aerosol Technik will not be liable for any damages caused by misuse of patent rights or rights against the third parties.

The hardware and software are continuously improved and extended, therefore the descriptions and illustrations in the manual may vary from the versions available to you. Please request the current version of this document.



Do not operate the instrument until you have thoroughly studied the operating instructions!
The manufacturer is not liable for damages caused by improper installation and use, cleaning, or handling errors.

2.3 Warranty

GRIMM Aerosol Technik warrants that the instrument described in this document has been designed, constructed and manufactured in accordance with the existing technical requirements for the application described, is free from defects in materials and from assembly errors, due to strict quality control. No further warranty however is given for an application-specific function or damages resulting from material or assembly errors. The manufacturing process of each instrument is monitored and precisely recorded, in particular calibration and validation.

Should the device fail during the 1-year warranty period or fail to meet the requirements, Grimm Aerosol Technik has the right to replace the defective part(s) or the instrument, except in the case of operating errors. The Grimm Aerosol Technik company shall carry out the repair(s) in the factory free of charge, excluding the transport and associated costs to be borne by the customer. The on-site repairs shall be carried out only against reimbursement of the travel and service costs by the customer. Further claims, which may arise from the warranty, are not at the expense of Grimm Aerosol Technik.

GRIMM Aerosol Technik assumes the warranty of the sold goods only, if they are used under normal conditions and in accordance with the instructions in this manual. The warranty expires after the 12 month period, starting from the date of delivery from the factory. Returns for warranty works shall be at the expense of the customer.

This warranty is interrupted by the following exceptions:

- 90 days of liability for spare parts replaced or repaired during the warranty in order to restore the operation of the instrument under normal use only.
- The manufacturer is not be liable for any supplier products, batteries, or consumables. Only the original warranty remains valid.
- Without written confirmation from GRIMM Aerosol Technik, no warranty is given for supplier parts, which have been modified, disassembled or installed by untrained personnel.
- The above replaces all other previous warranty agreements or limitations. There are no further liability claims, in particular outside of the normal use.
- The buyer is responsible for the use and operation. He must comply with the legal requirements and conditions and operate the instrument in accordance with the legal and operational regulations. Any deviations herefrom shall lead to exclusion of warranty.
- Legal means against GRIMM Aerosol Technik, no matter from which side brought in, are invalid after 12 months.
- The buyer and the seller agree that this limitation of warranty, which constitutes the requirements and limits, shall not be called into question. Both partners hereto are fully qualified merchants under German law.
- In case of a dispute, the court of jurisdiction shall be Traunstein, Germany.

2.4 Safety requirements

The manufacturer declines any direct or indirect liability, if the instrument is opened or manipulated! To ensure safe operation, the user must follow the safety instructions in this manual.

If it can be assumed that safe operation is no longer possible, the instrument must be taken out of operation and secured against unintentional use. It can be assumed that safe operation is no longer possible if:

- the instrument is visibly damaged,
- the instrument no longer works,
- the instrument has been stored for a longer time under unfavourable conditions, or
- exposed to heavy use

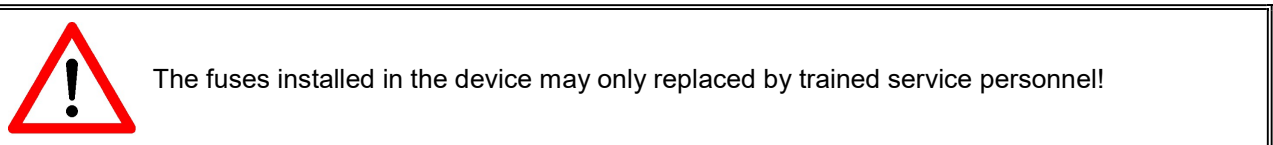
If the acclimatization time for a cold device, e.g. after transport at low temperature, is not observed before commissioning, the pump may block due to condensation. In this case, the electronic overcurrent fuse will completely switch off the device.

2.5 Electrical safety

The device meets the general requirements under the EMC Directive (EN 61326-1:2013), as well as the requirements for radio interference (en 55011:2011) and interference immunity (DIN en 61326-1:2013). Since the device is operated with the low voltage protection, a test according to DIN en 61010-1:2011-07 or VDE 0411-1:2011-07 is not required.

Before using the 11-D, check that the mains voltage is within the allowed range of the instrument. The wide range power supply is designed for AC voltages from 100 to 240 VAC at 47 to 63 Hz. The 11-D may only be operated on a circuit with an installed FI circuit breaker.

The instrument is protected by electronic overcurrent fuses. The fuses respond only to a serious fault.



2.6 Laser safety

In normal operation, the device meets the requirements for class 1 lasers according to DIN EN 60825-1: 2015. This is indicated by the information label on the back of the device; see Figure 2-1 on the left.

Class 1 lasers meet the highest protection class requirements so that no potentially hazardous radiation is accessible during normal operation.

When opened, the laser protection class 3B according to DIN EN 60825-1: 2015 applies. Class 3B lasers pose danger to eyes and skin. An appropriate triangular warning sign is attached to the laser holder inside the instrument; see Figure 2-1, in the middle. The information sign "WARNING – LASER RADIATION - – DO NOT EXPOSE TO THE BEAM, LASER CLASS 3B ", which is also required for class 3B lasers, cannot be practically installed inside the instrument due to the size and construction of the device. For this reason, the warning sign is listed in this manual, see Figure 2-1, right.

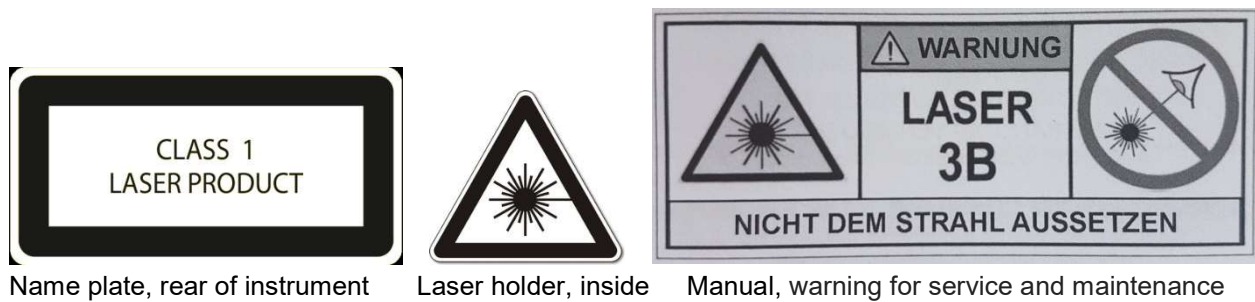


Figure 2-1: Laser Safety - Warnings

During service or repair work on the open device, if laser radiation is accessible, appropriate laser safety goggles or adjustment goggles must be worn. Further information on laser safety and possible dangers can be found e. g, for example, in the DIN EN 60825-1: 2015.

2.7 Applications and appropriate use

2.7.1 Applications and general information

The Aerosol Spectrometer Model 11-D is suitable for all types of aerosol measurements, provided that the specified requirements are met and the measurements are not made continuously or permanently outdoors. Such measurements include: determination of inhalable (E-dust) and respirable (A-dust) dust concentration at the workplace according to EN 481, measurement of the PM_{2.5}, concentration indoors according to VDI 4300-11, **testing and assessment of the absorption capacity of fine dust in sweepers according to EN 15429-3**, mobile measurements indoors or outdoors, determination of the particle size distribution, issues of aerosol research.

The 11-D is designed for portable, mobile use. In combination with an SMPS+C, the 11-D can also be operated as a wide-range aerosol spectrometer. This measuring system detects particles in the size range of 5 nm to 35.15 µm.

2.7.2 Automatic altitude correction

Depending on the air pressure, the sample volume flow is automatically regulated to the setpoint of 1.2 liters per minute. A manual correction of the volume flow as with the previous models is not necessary. Automatic altitude correction ensures high precision particle size detection for all locations, regardless of ambient pressure.

2.7.3 Information about operation and remote control

The 11-D can be operated via various interfaces such as RS-232, Ethernet, USB or wirelessly via Bluetooth. Operation and data transmission are only limited by the maximum cable length or the range of the Bluetooth radio connection.

2.7.4 Notes on starting and stopping the measuring operation

To start and stop the measuring operation, the 11-D must be supplied with voltage via battery or mains adapter. Approved mains voltages are documented in the technical specifications. The measurement starts when the instrument has run through the self-test. Manual starting or stopping the measuring operation is possible at any time. The device also allows start and stop times to program automatic measurements.

2.7.5 Memory functions by the mains voltage interruption

The 11-D has a memory function in case of power failure. If the 11-D is powered by the AC adapter without a battery and the power supply is interrupted, e.g. due to carelessness, the measurement is automatically resumed when the voltage is restored. The last type of display selected in standby mode and the alarm threshold remain stored in the device. The calculated filter weight and the sample volume are stored continuously.

2.7.6 Preparatory workflows and checks

Every new measurement starts with a self-test, which gives information about the instrument status. It is therefore obligatory for the 11-D to ask whether the filter has been changed. This filter is integrated in an accessible filter chamber in the unit and is usually changed by the customer only if a gravimetric validation of the fine dust calculation is to be carried out or the pressure drop through a heavily loaded filter becomes so high that the motor current for the pump increases disproportionately high. In addition, the filter is changed by the service personnel during the annual calibration and maintenance. Therefore, the filter query can usually be answered with no {-}. If the filter query remains unanswered on the display for a certain time, the instrument assumes that the filter has not been changed and switches to standby mode.

2.8 Application restrictions

To ensure that the 11-D functions correctly, the conditions for temperature, relative humidity and ambient pressure, listed in Table 1 in Chapter 3, must be observed.

If the particle number concentration is too high, coincidence occurs. Coincidence is the simultaneous occurrence of two and more particles in the beam passage. In such a case, several particles are counted as one particle with too large diameter. At a particle concentration of more than 3,000,000 particles / liter, a suitable diluent should be used.

In the case of high humidity or when measuring droplet aerosols, there is a risk that moisture will condense in the instrument. This can cause faulty measurements or damage to the device. To avoid this, make sure that the device is not colder than the ambient air. A sufficient acclimatization time and the values of the temperature and humidity sensor in the measuring cell must be observed in such cases.

In case of doubt and questions concerning the use of the fine dust measuring instrument model 11-D, please contact your dealer or the Grimm Aerosol Technik sales representative.

2.9 Transport

For transportation the 11-D must be switched off and can be transported only in its original packaging. The aerosol inlet must be closed with the supplied cap.

2.10 Storing

The original packaging should be used to store the instrument, and the temperature and humidity specified in chapter 3 for storage and transport should be observed. During storage, the aerosol inlet must be closed with the supplied cap.

Please inspect the individual parts of the measuring system after unpacking. If it can be assumed that safe operation is no longer possible, the instrument must not be operated and must be secured against unintentional use.

2.11 Disposal

2.11.1 Recommendations for the disposal of waste

The used PTFE filters can be disposed of with the household waste. If during application special aerosols were produced and measured with the 11-D, e.g. active substances, toxic or radioactive particles, the contaminated PTFE filters must be disposed of appropriately.

The Li-ion battery used in the 11-D complies with the requirements of EU RoHS2, China RoHS and EU REACH. If a battery can no longer be used, it must be disposed of properly.

2.11.2 Expected service life of the instrument

If used as directed, serviced regularly or serviced by an authorized service technicians, the Model 11-D has a lifetime of many years. It is not possible to make a statement about a standard lifetime, which is subject to the respective application.

2.12 Repair

The policy of the house GRIMM is to react immediately to any problems reported by the customer, pertaining to the defective or inactive instruments. If a device standstill / failure is detected, please contact the nearest GRIMM sales office or dealer immediately.

Prior to sending any of our devices for service, please contact the service department of Grimm Aerosol Technik by email, at:

service@grimm.durag.com

and specify the following data:

- Instrument model (number)
- Serial number and year of manufacture (on the name plate on the back of the device)
- Order date and your order number (except for a warranty claim)
- Your billing address
- Your shipping address



Make sure that the device is free of harmful contaminations before shipping the device for service and send the completed Declaration of Contamination with it.

3 Technical data

3.1 Measuring principle

The Aerosol Spectrometer Model 11-D is a portable instrument designed to measure dust in air and its particle size distribution. The following measured values can be displayed or output: dust mass fractions TSP, PM₁₀, PM₄, PM_{2.5}, PM₁, PM_{coarse}, inhalable, thoracic, respirable dust mass fraction, total particle concentration (TC) and particle size distribution in 31 channels. The measured values are output as a mass concentration with the unit $\mu\text{g} / \text{m}^3$, or as a particle number concentration in particles / liter. The air is drawn in via a radially symmetrical suction head and directed straight into an optical measuring cell.

The instrument works on the principle of scattered light measurement on individual particles. The light source is a laser diode. The light, as a narrow laser band, is focused onto the measuring cell and directed into a light trap at the end of the measuring field. The entire sample air is aerodynamically focused into the focal point of the laser band, according to Annex A.3 "Definition of the sensing zone", ISO 21501-1: 2009-06. When a particle passes through the laser beam, it is illuminated. This results in scattered light. The scattered light is directed via a wide-range optical system to a receiver diode. The position of the detector is perpendicular to the laser beam. Each electrical signal of the diode is counted and classified into 2 x 16 size channels after an appropriate amplification. This results in a total of 31 size channels, since the size channel $> 2.982 \mu\text{m}$ is measured twice.

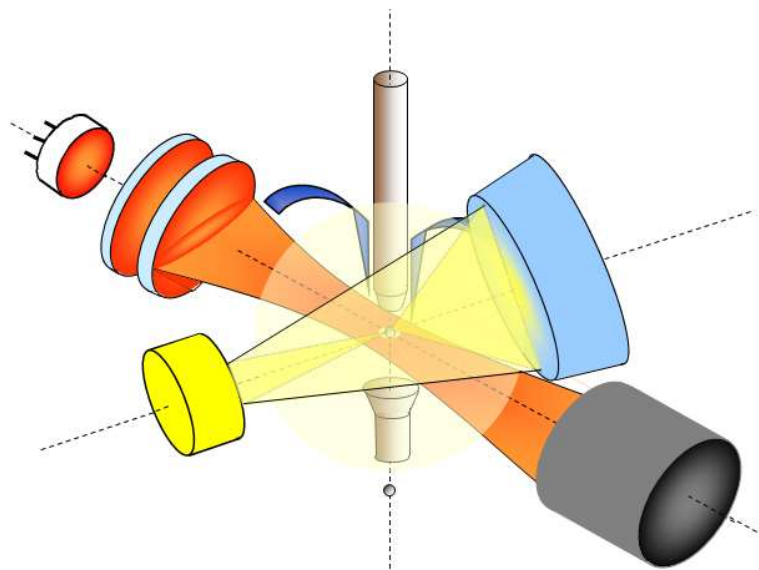


Figure 3-1: Schematic structure of the optical measuring cell

The calculation of the dust mass fractions is based on the measured number and size distribution. Assuming spherical particles, the volume distribution is calculated from the measured size distribution and, in a further step, the mass distribution is calculated by multiplication with the size-dependent density factors. When calculating the PM fractions, the size-dependent density factors and results from comparative measurements in ambient air with gravimetric dust collectors are considered. The three occupational medical dust mass fractions and the three pm-fractions are calculated with a uniform density factor for all size channels. This factor can be validated with an integrated PTFE filter.

The sample volume flow is set automatically. All particles contained in the aerosol stream are separated in a filter downstream of the measuring cell. The sample air pump conveys also the rinsing air, obtained via a micro filter from the pump exhaust air in the unit, and kept constant by a rinsing air controller. The rinsing air prevents contamination of the laser optics and the measuring chamber, and is used as particle-free reference air in the self-test of the instrument.

3.2 Technical data

Table 1: Technical data of 11-D

Specifications	Description
Detection principle	Light scattering at single particles
Light source	Durable diode laser
Volume flow rate	1.2 L/min, $\pm 3\%$ constant due to self-regulation, according to ISO 21501-1
Rinsing air flow rate	0.3- 0.5 L/min
Size channels	31, equidistant in μm : 0,253/ 0,298/ 0,352/ 0,414/ 0,488/ 0,576/ 0,679/ 0,800/ 0,943/ 1,112/ 1,310/ 1,545/ 1,821/ 2,146/ 2,530/ 2,982/ 3,515/ 4,144/ 4,885/ 5,758/ 6,787/ 8,001/ 9,431/ 11,12/ 13,10/ 15,45/ 18,21/ 21,46/ 25,30/ 29,82/ 35,15
Mass calculation	PM values: optimized for best correlation with the reference method, namely filter collector with gravimetric mass determination Dust masses according to EN 481 (inhalable, thoracic, respirable) with uniform density in all size channels pm-fractions for indoor or technical applications (pm_{10} , $\text{pm}_{2.5}$, pm_1) with uniform density in all size channels
Mass concentration	0 – 100.000 $\mu\text{g}/\text{m}^3$, depending on the aerosol (density)
Number concentration	0 – 3.000.000 particel/Liter
Data output	TSP, PM_{10} , PM_4 , $\text{PM}_{2.5}$, PM_1 , $\text{PM}_{\text{coarse}}$, inhalable, thoracic, respirable, pm_{10} , $\text{pm}_{2.5}$, pm_1 , TC, and particle concentration for all size channels on the display, via interface or USB flash drive
Reproducibility	more than 95% at 500 nm meets ISO 21501-1
Time resolution	6 seconds (31 size channels 0.253 μm – 35.15 μm) 1 second (16 size channels 0.253 μm – 2.982 μm , laser high) 1 second (16 size channels 2.982 μm – 35.12 μm , laser low)
Storage intervals	selectable, 6 seconds, 1, 5, 10, 15, 30, 60 minutes or 1 second
Data interfaces	serial: RS-232 with special Grimm data cable, USB (Type-B), Bluetooth, Ethernet: patch and crossover cables are possible
Data protocol	ASCII characters
Self-diagnosis	at every start and automatically when Autocal > 0
Operation	via membrane keyboard or PC via data interface
OLED-Display	2 * 16 alphanumeric characters, illuminated data output OLED display: dust measured values as moving average over one minute or averages with associated sample volume, alarm threshold, date, time
Analog inputs	3 units (0-10 V), 10-bit resolution (approx. 10 mV), display with editable texts, raw sensor data (analog voltage) are converted into measured values using calibration factors (offset and slope)
Data storage	USB flash drive. Dust measurement values, date, time, location number, error code, pump current, sensor data, calculated filter weight with sample volume, GPS coordinates, temperature and relative humidity in the measuring cell
Power supply	wide range power supply 100 to 240 VAC at 47 to 63 Hz, out: 13 VDC, 2.5 Li-ion battery, 10.8 V, 6.8 Ah, 73 Wh
Power consumption	5.4 W standard
Operating temperature	+4°C to +40°C, r.F. < 95 % (non-condensing)
Transport and storage	-20°C to +50°C, r.F. < 95 % (non- condensing)
Pressure	533 – 1133 hPa
Dimensions	281.5 * 123.5 * 67 mm (11.1 x 4.9 x 2.6 in)
Weight	1.6 kg (3.5 lbs), Li-ion battery 0.33 kg (0.7 lbs)

3.3 Scope of delivery and optional accessories

3.3.1 Scope of delivery

The following list contains the scope of delivery for the 11-D (set) with item number and product name.

Table 2: Scope of delivery 11-D (set) accessories

Item No.	Product name	Number figure 3 2
4021773	11-D Spectrometer	without picture
4021280	Li-ion battery	2
4022459	Charger for Li-ion battery	1
4022554	Power supply 13 VDC	3
1200017	Radial symmetric sampling head	6
1200018	Connecting pipes	5
1200020	Minifilter for 0-Test	4
1200069	PTFE filter (25 pcs.)	7
1200073	RS-232 to USB cable	10
1201651	USB flash drive	9
4024240	Case 11-D	without picture
1201650	USB flash drive 2GB with PDF manual, software and driver	8
1118D	Digital manual	
1179	Spectrometer-Software	

Figure 3-2 shows the scope of delivery of accessories, without case and country-specific power plug.



Figure 3-2: Scope of delivery 11-D (set) accessories

3.3.2 Optional accessories

The following list contains optional accessories for the 11-D with item number and product name.

Table 3: Optional accessories

Item No.	Product name
1200019	GPS receiver
1200021	Isokinetic sampling set
1200084	Sensor for temperature and relative humidity
4019778	Soft carrying bag with shoulder strap
4019167	Diluter 1:10
4018201	Diluter 1:100
4022500	Power cable 250 VAC 10A C13 / UK mains. 1.8m
1200657	Power cable 250 VAC 10A C13/US mains. 1.8m
4022460	Charger for 2 Li-Ion batteries with calibration function



Carrying bag, GPS receiver, T rF sensor



Isokinetic sampling set



Diluter 1:10 (Diluter 1:100)



Charger for 2 Li-ion batteries with calibration function

Figure 3-3: Optional accessories

4 Instrument description

4.1 Operating controls

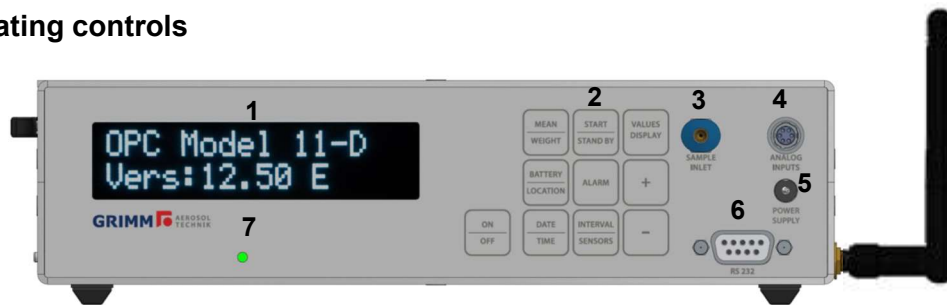


Figure 4-1: Front view



Figure 4-2: Rear view

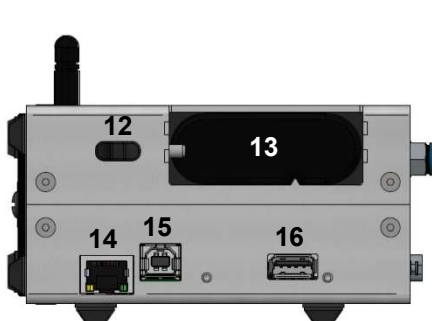


Figure 4-3: Side view left

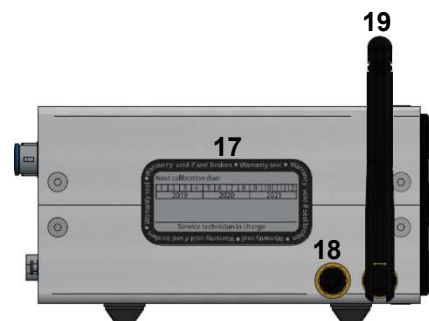


Figure 4-4: Side view right

The following descriptions of the controls refer to numbering in the pictures 4-1 to 4-4.

Table 4: Controls

Numbering	Description
Front side	
1	Display
2	Keyboard with on-off button
3	Sample air inlet
4	Analog input
5	Socket for power supply (power supply)
6	RS-232
7	Status LED
Rear side	
8	Filter compartment with bayonet lock
9	Tripod socket ¼ "- 20 UNC - 2A
10	Name plate
11	Sample air outlet
Left side	
12	Lock button for Li-ion battery
13	Battery slot
14	Ethernet socket
15	USB socket (type B)
16	Socket USB flash drive (Type-A)
Right side	
17	Calibration sticker and security label
18	GPS connection (female)
19	Bluetooth connection (male) with antenna

4.2 Display

The optical display of the device consists of an illuminated OLED display with a total of 2 x 16 characters. The display shows the measured values, date, time, measuring location number and any messages that occur.

4.3 On / off switch, standby and measurement mode

The 11-D has three operating states: switched off, switched on in standby and switched on in measuring mode. The **{ON / OFF}** key turns the device on or off. The prerequisite is that the unit is connected to the mains adapter or that the battery is locked in the battery compartment. The instrument also turns on automatically, when connected to the AC adapter, or when the battery is inserted into the battery compartment and locked.

After switching on, the status LED lights up blue and then changes to green. The display is activated.

The device must never be switched off directly from the measuring mode; it should always be switched to standby by pressing the **{START / STAND-BY}** key. Non-compliance can, in rare cases, result in data loss. The instrument interprets switching off in measuring mode as a power failure; the measurement is automatically continued upon restarting, without a new filter query and with the mean values of the previous measurement. After switching on, the status LED lights up once in blue and then changes to green. The display is activated.

4.4 Keyboard

The instrument has ten keys for operation. Some of the keys have different functions in measuring mode and standby. When the keys are pressed, a short confirmation beep sounds.

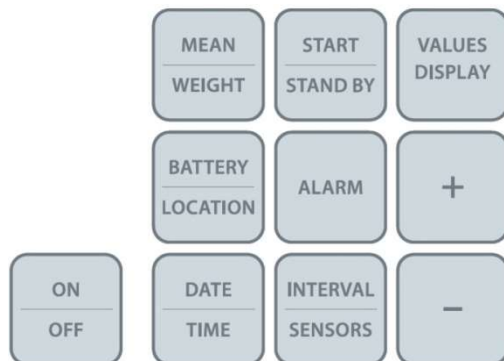


Figure 4-5: Keyboard

	The {ON / OFF} key turns the unit on or off. The button must be pressed until a beep sounds. While the measurement is running, the device must not be switched off. First, the measurement must be ended properly by pressing {START / STAND BY} .
---	--

4.4.1 Key functions before response to filter change

The instrument 11-D is switched on - no previous power failure noticed.

Table 5: Key functions before response to filter change

	Display of the filter weight (Weight) with the corresponding sample volume (F-Vol). The values remain displayed as long as the key is pressed. All measurements after the filter change are included in the calculation. The filter weight is calculated allways with a default gravimetric factor 1.00.
	Display of serial number and operating time.
	Change the storage interval with {+} and {-} . Note: The storage interval cannot be changed during standby or during measurement.
	"Yes", filter was changed. The filter weight and the average values are deleted. The unit starts with a self-test.
	"No", filter was not changed. The filter weight and mean values are not deleted. The unit starts with a self-test.

All other buttons work as in standby.

4.4.2 Key functions in standby

Table 6: Key functions in standby

	Display of the mean value of the selected measured values (particle concentration and / or dust mass fractions) calculated from all measurements after switching on the instrument. The values remain displayed as long as the button is pressed. The sample volume (M-Vol) appears after releasing the button in the lower display line and remains displayed for approx. two seconds.
	Pressing {MEAN / WEIGHT} and the key {-} simultaneously for 5 seconds, the mean value, and the sample volume is set to zero.
	Pressing the button will leave the standby and start a new measurement. The self-test follows. The device is then in the measuring mode.
	After pressing the button, you can select two measurement values to be displayed in the display by pressing the {+} or {-} key. The following measurement values are possible: concentration of dust mass TSP, PM₁₀, PM₄, PM_{2.5}, PM₁, PM_{Coarse}, respirable, thoracic, inhalable, pm₁₀, pm_{2.5}, pm₁, TC or cumulative particle number concentration from a selected lower channel limit.
	Pressing the key displays the current battery capacity in % and the thereof resulting remaining measurement duration in hours, as well as the set location number. When the AC adapter is connected, "External Power" appears in the display. By pressing and holding the key {BATTERY / LOCATION}, the location number can be changed with the {+} or {-} buttons. The location number is used to differentiate between various measurements and can be set between 1 and 99.
	By holding down the key, you can use {+} or {-} to set a threshold at which an alarm is triggered. "0" means that the alarm function is switched off. The threshold refers to the measurand displayed in the top line of the display. The alarm function can therefore be used for the measured particle number concentration or dust mass
	Short Press: Display of date and time. Longer press: Set the date and time. The location of the blinking cursor can be changed by pressing the {+} or {-} key. By pressing the {DATE / TIME} button again, the cursor is switched on. If no button is pressed for a long time, the return to standby is performed. When you change the minutes, the seconds are set to zero.
	Changing the input values in combination with other buttons {VALUES / DISPLAY}, {BATTERY / LOCATION}, {ALARM}, and {DATE / TIME} or parameter reset with {MEAN / WEIGHT}.

	Pressing the button displays the current values of connected sensor (temperature and relative humidity) in the display (upper line) and the time (lower line). Note: in standby or during a measurement, the storage interval can not be changed with the keyboard, but only during the filter prompt "Filter changed? press +: yes -: no ".
	Press {ON / OFF} to turn the device on or off. The button must be pressed until a beep tone is heard. The unit must not be switched off when the measurement is running. Before this, the measurement must be terminated properly by pressing {START / STAND BY}.

4.4.3 Key functions in measuring mode

Table 7: Key functions in measuring mode

	Display of the mean value of the two selected measurement parameters (TSP, PM₁₀, PM₄, PM_{2.5}, PM₁, PM_{Coarse}, respirable, thoracic, inhalable, pm₁₀, pm_{2.5}, pm₁, TC or particle number concentration calculated from all measurements after switching on the device. After about 5 seconds, the sample volume corresponding to the mean values appears and remains displayed as long as the button is pressed.
	By pressing the {MEAN / WEIGHT} and {+} key simultaneously, the filter weight and the corresponding sample volume can be displayed. All measurements after the filter change are taken into account. The query of the filter weight provides information on whether the particle mass deposited on the filter can be detected by the available balance within the stated error limits during a gravimetric validation of the occupational-medical dust mass calculation or the pm values. All measurements after the filter change are included in the calculation. The filter weight is always calculated with the default gravimetric factor 1.00.
	Pressing this button stops the current measurement and switches the device to standby.
	A new value (moving mean) is displayed on the display every 6 seconds. If the {VALUES / DISPLAY} button is pressed briefly, the next measure is selected in the lower display line and the calculation of the moving mean values for both lines is restarted. If the key is held down, the measures in the lower display line change every second. Releasing the key will restart the calculation for both lines. The actual measurement remains unaffected.
	Pressing the button will display the current battery capacity [%] and the battery life [h] as well as the set measuring location number. "Power Supply" appears when the power supply is connected.

	Retrival of the currently set alarm threshold. A "0" before the unit $\mu\text{g}/\text{m}^3$ or P/liter means that the alarm function is switched off.
	Switching on the cyclic display of the sensor values. In the first line of the display, the sensor values are displayed in addition to the set measured value.
	Pressing the button stops the cyclic display of the sensor values.
	The current date and time are displayed when this key is pressed.
	As long as the button is held down, the current sensor values (e.g. temperature and relative humidity) are shown on the display. Note: in standby or during a measurement, the storage interval can not be changed with the keyboard, but only during the filter prompt "Filter changed? press +: yes -: no".

4.5 Sample air inlet

The sample air inlet on the front of the device is labelled "Sample Inlet". It consists of a 4 mm quick-release coupling (Festo) and fits all common GRIMM sampling systems (radial symmetric sampling head, connection tube, isokinetic channel probe) with 4 mm outer diameter.

Assembly: insert the sampling system into the sample air inlet until it locks. An internal sealing ring fixes the sampling system airtight.

Dismantling: To remove the sampling system, the blue plastic sleeve at the sample air inlet must be pressed downwards. This opens the latch and the sampling system can be easily pulled out.



Before removing the radially symmetrical sampling head, always press down the blue plastic sleeve at the "Sample Inlet". With other sampling systems (connection tubes, channel probe) proceed in the same way.

4.5.1 Radially symmetric sampling head

The radially symmetric sampling head of the 11-D is a TSP head. There is no size-selective deposition in the sampling head. For reproducible sampling, it is recommended to use the radially symmetric sampling head for all standard measurements, unless other sampling systems such as the isokinetic channel probe are used. The capillary diluter is also compatible with the radially symmetric sampling head.

4.6 Analog input

An optional sensor for temperature and relative humidity can be connected to the analog input on the front of the unit. This sensor is necessary for the calculation of PM_{10} , $\text{PM}_{2.5}$ and PM_1 values with SVC correction.

4.7 Socket for power supply unit

The socket for the power supply unit is located in the middle right of the front of the unit. For measurements without battery or continuous measurements over 10 hours, the power supply unit must be used. Unlike the previous models, the battery is not recharged in the device even when the mains adapter is plugged in.



A special power supply unit with protection class I is applied, which provides the protective extra-low voltage 13 VDC (PELV) on the output side. It is used exclusively for the power supply and grounding of the instrument. Due to the low voltage and the absence of an internal charging circuit, unwanted heating of the device is avoided.

For this reason, no 18 VDC power supply unit of the previous models or any other PSU for the 11-D should be used.

4.8 Status LEDs

The LED on the front panel indicates the following states:

- OFF
- Green bright instrument switched on in measuring mode or standby
- Blue bright Bluetooth connection active
- Turquoise bright firmware update (only by trained service technicians)

There is a red LED on the port for the USB flash drive on the left side, which flashes when data is written to the USB flash drive.

4.9 Filter compartment with bayonet lock



The device must never be operated without a filter. Even if no gravimetric correction of the calculated dust masses is made, a filter must be used in the filter compartment.

Without a filter, all particles in the sample air enter directly into a pump protection filter, which is installed between the filter compartment and the pump. This prevents the pump from being contaminated. The pump protection filter is dimensioned in such a way that it is exchanged once a year. Without filters in the filter compartment, the life of the pump protection filter is shortened considerably. Due to the increased pressure loss the pump wears out faster.

4.10 Tripod socket

With the tripod socket on the back of the unit, the 11-D can be mounted on a tripod. As a result, measurements under reproducible conditions (e.g. defined height) can be quickly and easily performed.

4.11 Sample air outlet

The sample air outlet on the rear of the unit is protected against contamination by a black plastic cover, which must be removed to install the isokinetic channel probe. For all other applications, the cover can remain on the device.

4.12 Battery compartment

The battery compartment is located on the left side of the instrument. It is designed so that the Li-ion battery can only be inserted in the correct orientation. The battery must be then secured with the locking knob. A full battery allows 10 hours or more of measuring operation.



The Li-ion battery cannot be charged in the device. To charge the battery, remove it from the 11-D and use the separate desktop charger.

4.13 GPS connection

For position information in mobile measurements, the 11-D can be connected to a GPS receiver. The connector is located to the right of the Bluetooth antenna. To avoid confusion, the GPS connection is female, the connection for the Bluetooth antenna male.

4.14 Bluetooth antenna

The 11-D is equipped with Bluetooth allowing a wireless connection to the measuring computer. The antenna can be turned and tilted to optimize the signal strength. When delivered, the Bluetooth antenna is pre-assembled, but can be easily dismantled. When mounting, pay attention to the correct connection, see Figure 4-4: Side view right.

4.15 Name plate, calibration sticker and security label

The name plate on the back of the device contains all relevant instrument information and safety instructions

- Manufacturer's instructions
- Model description
- Serial number
- Bluetooth name
- Production year
- Operating voltage and polarity of the power supply
- Protection class
- Caution for laser safety

The calibration sticker on the right side of the device is designed as a safety label. It informs about the time of the next calibration and is destroyed by unauthorized opening of the device. A device with a damaged or removed calibration sticker loses any warranty claim.

4.16 Data storage / memory with USB flash drive

A USB flash drive can be used to store without a connected PC. The USB flash drive is plugged into the left-hand side of the battery compartment.



The USB flash drive must not be plugged in or removed from the device when the measurement is running. It must not be removed until the red led flashes.

In the 11-D, USB flash drive with more than 4GB can be used. Even with a 6-second measurement interval, the storage capacity lasts several months. However, the memory interval should be adjusted to the duration of the measurement. Days of continuous measurements in the 6-second interval are not useful and can lead to files that are no longer readable due to their size (e.g. Excel).

The file name of a measurement is composed of a two-digit indication of the location and a four-digit run number, separated by an underscore. Example: 01_0001.txt



Some important hints, which ensure the trouble-free handling of the USB flash drive.

- The USB flash drive may only be changed in standby mode or when switched off, as this can result in data loss.
- If the USB flash drive has not been inserted before starting measurement, the instrument must be put into standby mode by pressing the **{START / STAND BY}** key and then the USB flash drive can be inserted.
- The USB flash drive cannot be read out or deleted in the 11-D. It must be plugged into a PC with a USB interface and can then be read and deleted from the PC.
- The USB flash drive must be formatted in the format Fat 32.
- It is recommended to delete the unnecessary files from the USB flash drive. If the USB flash drive is full, no further measured values can be stored. The device gives no warning in this case.

4.17 Data interfaces

The 11-D has the following interfaces: RS-232, USB (Type-B), Ethernet and Bluetooth. Via the interface the communication between 11-D and a PC can be established. The position of the sockets are marked in Figure 4-3: Side view left. For wired communication, the device must be connected to a PC via the corresponding socket and the associated cable.

4.17.1 RS-232

For an RS-232 connection the data cable (article number 1200073) is mandatory, because the RS-232 socket has a special pin assignment. The data cable is provided with an interface converter RS-232 to USB (Type A), see No. 10, Figure 3-2.



Figure 4-6: Pin assignment of the "RS-232" socket on the front of the aerosol spectrometer



Because of the alarm output on the 9-pin RS-232 subminiature socket, only use the original Grimm data cable.

4.17.2 USB (Type-B)

For connection via USB (Type-B) a corresponding standard cable can be used. The socket is located on the left side of the device, see No. 15 in Figure 4-3.



USB (Type-B) cable is not included in delivery as it is a standard cable. The customer can use any suitable USB (Type-B) cable.

4.17.3 Ethernet (LAN)

At the factory, the unit is provided with a preset IP address (192.168.11.0, network for private use RFC 1918), which is adaptable for use on a network by the customer according to his needs. The adaptation of the IP address can be done via the software or via HyperTerminal using the command **{\}** (backslash key), see chapter 6.3.2.

4.17.4 Bluetooth

At the factory, the device is equipped with a Bluetooth module. It allows a wireless connection over several meters. The connection setting must be carried out via the PC. The Bluetooth name for identification of the GRIMM device is affixed to the name plate under the name BT-Name and is identical to the serial number; see point 9 in Figure 4-2.

5 Start-up and measurement procedure

Before starting the measurement, it is necessary to determine which way the measured data is to be recorded and further processed. The following options are available:

- Saving the data on a USB flash drive without online data acquisition
- Online data acquisition on PC via RS-232 cable, USB cable, Ethernet or Bluetooth with or without simultaneous storage on a USB flash drive

The USBflash drive must be plugged into the 11-D before starting the measurements. Please note the information in chapter 4.16 for the installation and operation of the USB flash drive. After switching on the device, the status LED flashes blue once and then changes to green. The display shows the model name and the firmware version number "OPC Model 11-D, Version 12.50 E". After approx. 5 seconds, the date and time are displayed.

Then the filter query appears. During the filter query, some special functions are possible, as described in chapter 4.4.1. To prevent accidental deletion of the calculated filter weight and the corresponding sample volume, a delayed keyboard query is performed. The button must be pressed for at least one second until a beep sounds. When the filter query is complete, the measurement starts with the self-test. If the filter query remains unanswered, it disappears after about 10 seconds and the meter switches to standby. To start a measurement from the standby mode, press the **{STAR / STAND BY}** key.

Each measurement is initiated by a self-test, in which the measuring cell is rinsed with clean air. The self-test takes about 30 seconds, during various status values of the device condition are recorded. After completing the self-test, the message "Self Test OK" appears on the display.

After the self-test, the actual dust measurement begins. The moving mean value of the last minute is displayed on the display, calculated over 10 individual measurements. The measured value on the display is always updated after six seconds.

5.1 Measured values on the display

In standby (and in measurement mode), the display mode can be changed by pressing the **{VALUES / DISPLAY}** and **{+}** or **{-}** keys. Only two measurement values can be displayed at the same time on the two-line display. In contrast to all previous models, the 11-D can display both dust mass

values (mass) and number concentrations (counts) simultaneously and change the measured value display in the measurement mode.

In measurement mode, the values of the connected climate sensor can also be displayed cyclically and alternating in the first line by pressing the **{+}** key (see chapter 4.4.3).



In 6 second interval the values are displayed as sliding averages calculated over a period of one minute. The values are refreshed every 6 seconds.

5.1.1 Warning and error messages on the display

The operating status of the measuring instrument is continuously monitored electronically. If the set alarm threshold is exceeded and the device faults occur, for example a defect in the volume flow control, a message is output on the display and a warning signal sounds.

Table 8: Warning and error messages on the display

Alarm / error message	Meanning
AIRFLOW NOT OK! CHECK AIR INLET	The volume flow control cannot keep the sample volume flow constant at 1.2 L / min. The measurement stops. Action: Check the sampling head for contamination and clean if necessary.
PUMP CURRENT TOO HIGH	Power consumption of the pump too high. Action: Open the filter chamber and remove the gravimetric filter. Clean the sampling section with particle-free compressed air with max. 3 bar pressure from the front (sample inlet) in the direction of the filter chamber. Never vice versa. Optionally, use a new gravimetric filter. If the problem persists, contact Grimm service.
BATTERY LOW PLEASE RECHARGE	Battery capacity < 10 %. Action: If necessary, switch to mains operation to continue the measurement without interruption. Remove the battery and recharge it in the desktop charger (approx. 3 hours).
NEW SELF TEST	This error can be triggered in the self-test phase of the dust measuring device by various causes, for example: Device temperature over 40° C Very contaminated measuring cell Defect laser or other hardware failure Foreign matter stucked in the inlet nozzle, protruding into the laser beam The device repeats the self-test.
SELF TEST NOT OK	This error can have different causes. Action: Connect the device to a PC, establish a HyperTerminal connection, switch the device to service mode 0 (as described in chapter 8.2) and restart. Log the error message that is output via the interface and contact Grimm service for further action.
FATAL ERROR PLEASE CHECK	The repeated self-test failed. Action: Contact Grimm service.

6 Online data acquisition

All 11-D devices are powered by the same firmware version 12.50. There are no model variants. The user has two data protocols available: standard or Modbus.

The spectrometer software and HyperTerminal access the standard protocol. Modbus uses its own protocol.

6.1 Spectrometer Software 1179

The spectrometer software 1179 is compatible with 32- and 64-bit Windows operating systems, from Windows 7.

In 11-D, the measured values are displayed numerically or graphically in the following formats:

- Dust mass fractions TSP, PM₁₀, PM₄, PM_{2.5}, PM₁, PM_{Coarse}, inhalable, thoracic, respirable, pm₁₀, pm_{2.5}, pm₁, in the unit µg / m³ and TC in particles / liter
- Particle concentration for all channels in the unit of particles / liter
- Measurements of external sensors for temperature and relative humidity
- Service data, such as pump current, temperature and relative humidity in the measuring cell, i.a.

The 1179 spectrometer software enables online statistical data analysis as well as analysis of the device functions and complete system diagnostics.

6.1.1 Installation of the spectrometer software 1179

The software is provided on the supplied USB flash drive. The installation instructions and a license file (*.lic, e.g. 11D18001.lic) are included on the USB flash drive. The license file must be copied into the directory with the Spectrometer_1179_V*.exe file. The default directory for installation is: C:\Grimm\Spectrometer

Alternatively, the installation instructions and the software can be downloaded from the homepage of the Grimm Aerosol Technik Ainring GmbH & Co. KG. Also when installing from the homepage, the license file must be copied into the directory with the Spectrometer_1179_V*.exe file.

6.1.2 Screenshots of the spectrometer software

There are a few screenshots below to help you get started with the spectrometer software 1179. In order to make the use intuitive, the keys that allow a function are highlighted in light gray and the keys that do not allow a function in dark blue.

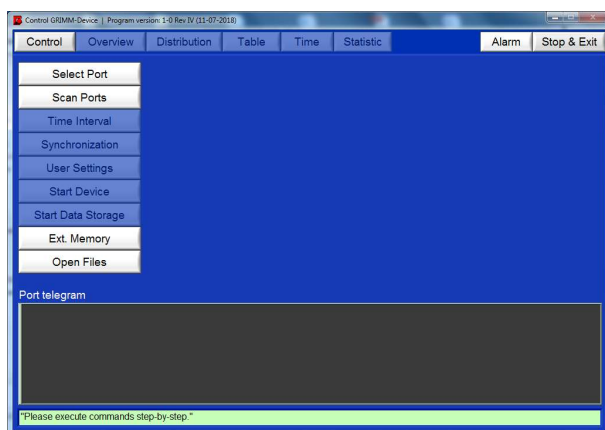


Figure 6-1: Start screen

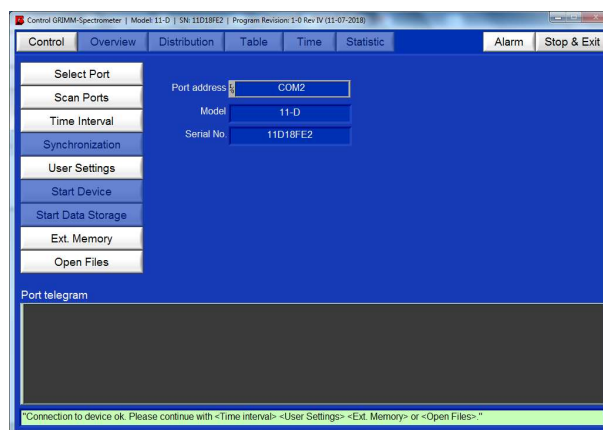


Figure 6-2: Selection of the interface

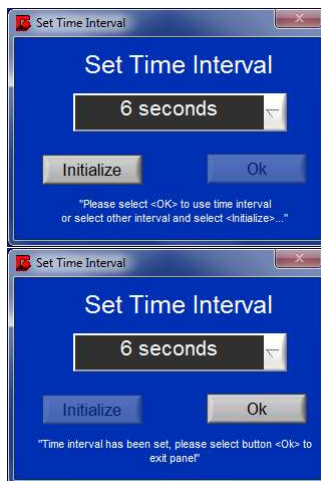


Figure 6-3: Selection and initialization of the storage interval in the Time Interval menu.

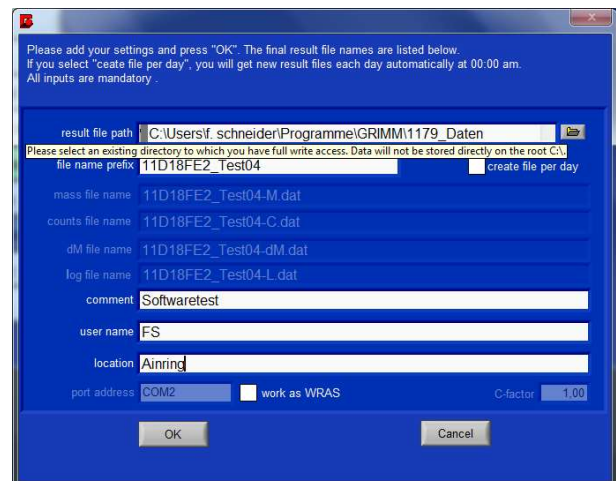


Figure 6-4: User Settings Menu

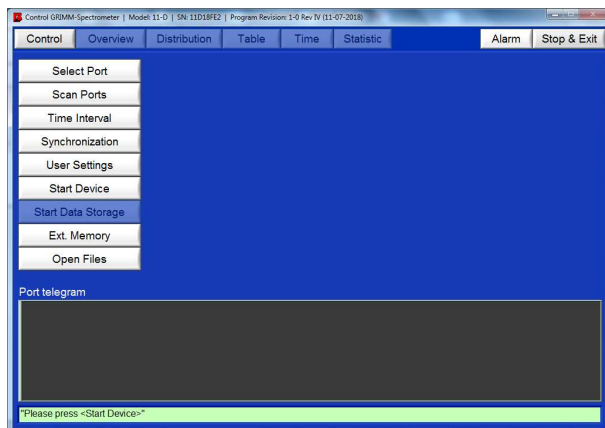


Figure 6-5: In the control window all menu items are active and highlighted in light gray

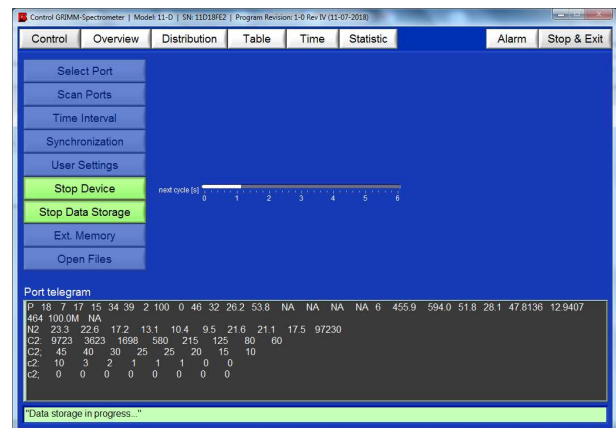


Figure 6-6: Initialization of the measurement by pressing "Start Device"

The measurement and recording of the measured values can now be started in the selection control by pressing {Start Device} and {Start Data Storage}. All other menu windows and chart options are now active.

The measured values are displayed and updated during the current measurement. Different chart options such as time history, size distribution, mass distribution or numerical values in tabular form are available for display. The software allows you to export the completed series of measurements as ASCII tables in EXCEL, as well as to open and display the completed series of measurements.

6.2 Modbus

The Modbus TCP implementation in 11-D has the following features:

- Access only via the RJ-45 socket on the left side of the device (Ethernet)
- IP address corresponds to the IP address set on the device
- TCP listener fixed on port 502 ("well-known-port" for Modbus TCP)

There are two data types available for the registers:

- Float: 32-bit IEEE 754 floating-point number, 1-bit sign, 8-bit exponent, 23-bit mantissa
- Unsigned: 32 bit unsigned, integer

One value occupies always 2 Modbus registers. The upper 16 bits of a value are stored in the respective start register of the measured value (see Table 9: Overview of the supported function codes). The lower 16 bits in the following register.

[illegible]

Register assignment for Float

[illegible]

Register assignment for Unsigned

Supported function codes

The 11-D provides the following function codes, where no functionality is implemented for the function codes 0x01 (read coils) and 0x0f (write coils). The corresponding coils are reserved for later possible extensions:

Table 9: Overview of the supported function codes

Name	Code
Read Coils	0x01
Read Holding Registers	0x03
Write Coils	0x0f
Write Holding Registers	0x10
Read Device ID	0x2b

The following Table 10 contains the assignment of the holding register when the interface is output via Modbus. Column 1 contains the name of the measured values, column 2 their respective unit, column 3 the start register, column 4 the file type signed (F = float) or unsigned integer (n = unsigned) and column 5 the read access (R = read) or read and write (R / W = read and write).

The measured particle number concentrations and calculated dust mass fractions are stored in the start registers 0 to 90. The lower channel threshold of the 31 size channels follows in the start registers 204 to 264. The registers 92 to 202 and 266 to 348 are not occupied.

The start registers 350 to 382 contain measured values of the internal and external (optional) sensors. This information varies depending on which optional sensors (e.g. GPS) are used.

Start registers 400 to 412 contain information on the status of the measuring cell, see K-line chapter 8.1.2. The start registers 420 until 426, 504, and 996 provide information about the device status. The meaning of the error codes is explained in chapter 6.3.2. A value of 0 means no error. Start register 998 contains the timestamp for the last update of the measured values as a Unix timestamp.

Table 10: Assignment of holding registers in Modbus

Measured value	Unit	Start register	Data type	Access
TSP	µg/m ³	0	F	R
PM ₁₀	µg/m ³	2	F	R
PM ₄	µg/m ³	4	F	R
PM _{2.5}	µg/m ³	6	F	R
PM ₁	µg/m ³	8	F	R
PM _{coarse}	µg/m ³	10	F	R
Total Counts	P/L	14	U	R
Inhalable	µg/m ³	16	F	R
Thoracic	µg/m ³	18	F	R
Respirable	µg/m ³	20	F	R
pm ₁₀	µg/m ³	22	F	R
pm _{2.5}	µg/m ³	24	F	R
pm ₁	µg/m ³	26	F	R
Particel number concentration Channel 1	P/L	30	U	R
...	...	...	...	...
Particle number concentration channel 31	P/L	90	U	R
Channel threshold channel 1	µm	204	F	R
...	...	...	...	...
Channel threshold channel 31	µm	264	F	R
Temperature (external)	°C	350	F	R
Humidity (external)	%	352	F	R
Wind speed	m/s	354	F	R
Air pressure	hPa	356	F	R
Degree of latitude	°	370	F	R
	(decimal)			
Degree of longitude	°	372	F	R
	(decimal)			
Hight	m	374	F	R
Temperature (internal)	°C	380	F	R
Feuchtigkeit (internal)	%	382	F	R
DC /V	mV	400	F	R
DC_d	mV	402	F	R
DC_h	mV	404	F	R
C0_h	P	406	U	R
C0_d	P	408	U	R
La_l	mA	410	F	R
La_h	mA	412	F	R
P-Weight	µg	420	F	R
P-Volume	m ³	422	F	R
Motor current	mA	424	F	R
Battery charge	%	426	U	R
Error code	-	504	U	R
Time self-test	s	996	U	R
Time measured values	s	998	U	R
Interval	-	1000	U	R/W
Operating status	-	1002	U	R/W
C-Faktor	-	1004	U	R/W

The coding for interval (start register 1000) and operating status (start register 1002) can be found in the following tables. The C-factor (start register 1004) can be set between 0.1 and 9.8 in steps of 0.1. The default value is 1.0.

Table 11: Coding of the interval

Content	0	1	2	3	4	5	6	7	8
Interval	1 Min	5 Min	10 Min	15 Min	30 Min	60 Min	6 Sek	1 Sek LH	1 Sek LL

Table 12: Coding of the operating status

Content	0	1	2
Status	Standby	Self-test	Measurement operation

6.3 HyperTerminal

Each Grimm spectrometer can be operated via HyperTerminal online with the appropriate control commands. HyperTerminal is a text-based communication program that has been included in the Windows operating system since Windows 2.0. As of Windows Vista, HyperTerminal is no longer included, but can be added from the pages of the company Hilgraeve as a paid or free download.

6.3.1 Establishing a wired HyperTerminal connection

The measuring instrument must be connected to a PC using an interface described in section 4.17. In Windows 7 you can find the program HyperTerminal under:

START / PROGRAMME / ZUBEHÖR / KOMMUNIKATION / HYPER TERMINAL

START / PROGRAMS / ACCESSORIES / COMMUNICATION / HYPER TERMINAL

1. Start HyperTerminal, enter name for new connection and confirm with OK.

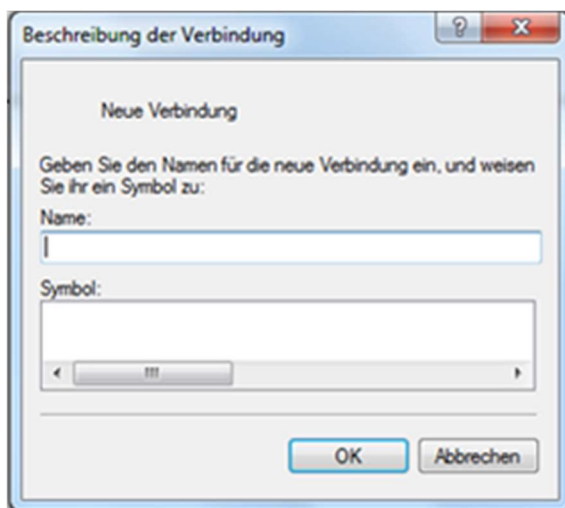


Figure 6-7: New HyperTerminal connection

2. Selection of the COMx or TCP / IP interface to which the device is connected. For a TCP / IP, the host address (IP address) and the port number 4711 must be additionally entered. Then confirm with OK. To change the IP address see chapter 6.3.2.

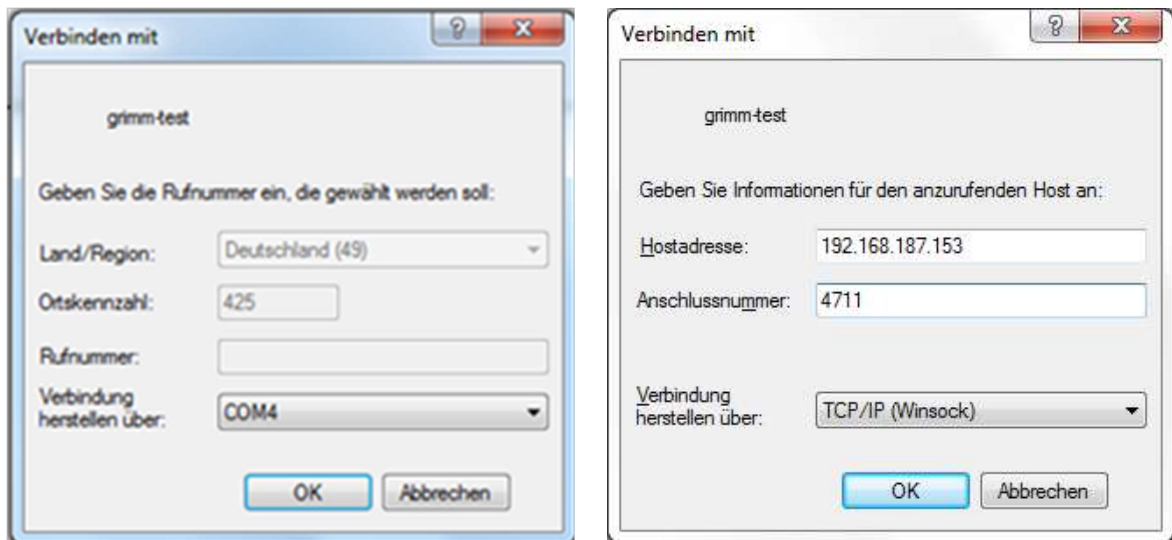


Figure 6-8: Interface selection

3. For a connection via a serial interface (COMx), the following settings must be made:

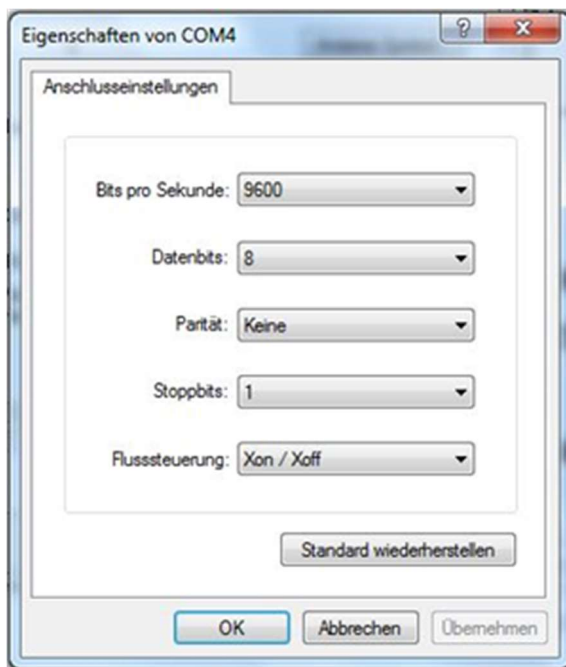


Figure 6-9: Transmission parameter

4. After the settings, the connection is checked by pressing 2 x {v}. Now, the version of the measuring device should be displayed on the PC.

```
v
Vers:12.50 E
```

The operation of the spectrometer is now possible with the control commands. {?} Opens the help window with an overview of all available commands. The commands are described in detail in chapter 8.2 RS-232 Commands.

6.3.2 Changing the IP Address

The IP address is preset for each device. It can be changed via HyperTerminal or via the spectrometer software. Changing the IP address with HyperTerminal:

- Switch the instrument to standby mode
- Connect the instrument to a PC with the RS-232 cable
- Start HyperTerminal on the PC
- By entering the \ (backslash), the IP address is displayed as follows.

\

IP:192.168.11.0 The upper output line shows the set IP address of the measuring instrument

IP:192.168.11.0 The lower output line is the editor line where the IP address can be changed.

The IP address can only be changed from left to right. All characters to the right of the cursor must be re-entered. The input must be confirmed by pressing ENTER (return key or enter key). After changing the IP address, the device will restart automatically. Finally, the active terminal connection must be terminated and the RS-232 disconnected.

7 Maintenance and cleaning

7.1 Maintenance and cleaning by the customer

Certain maintenance work on the 11-D can be carried out by the customer himself. Other maintenance work, such as internal filter replacement, must be performed by the authorized service technicians.



It is recommended to work through the following points at least once a year!

7.1.1 Cleaning the filter chamber

The filter chamber should be cleaned with a lint-free cloth or a soft brush each time the filter is changed. Stubborn dirt can also be removed with special cleaning cloths for EDP.

7.1.2 Cleaning the measuring chamber

The cleaning of the measuring chamber can be carried out only when the instrument is switched off. Open the filter chamber and remove the gravimetric filter. The air inlet on the front of the unit is equipped with the particle-free and oil-free compressed air which is blown at a pressure of max. 3 bar always in the direction of the air flow. Since the measuring cell flows in a straight direction from the sample inlet to the filter chamber, one must be able to see through the measuring chamber.

Contamination of the inlet nozzle should be removed from the filter chamber with a wooden toothpick (do not use harder materials to avoid damages of the inlet nozzle!).



Do not pull or remove any utensils (such as cleaning tissue, etc.) through the inlet nozzle!

7.1.3 Housing

The dust measuring device is surrounded by a metal housing that provides protection against mechanical influences and electromagnetic fields. The membrane keypad and the display must be protected from rough mechanical loads. In order to clean the housing, use a dry cloth or, for stubborn stains, damp special cleaning wipes for EDP equipment.



Protect the device against contact with liquids!

7.1.4 Replacing the gravimetric filter

The gravimetric filter (PTFE) must only be replaced when the device is switched off. By loosening the bayonet closure, the filter chamber cover can be opened without tools. The closure is provided with a fuse so that it can not fall out. Remove the used filter. Make sure that the dust collected on the filter does not get into the measuring cell through the nozzle on the filter chamber and contaminate it. If necessary, position the device so that the filter chamber can be opened sideways or downwards.

To clean the filter chamber use a commercial kitchen towel or a soft brush.

With tweezers, place a new filter in the center of the large O-ring of the filter chamber. The small O-ring should not be touched by the filter. Subsequently, close the filter chamber and lock with the bayonet lock up to a noticeable stop.



The maximum battery life, but also the service life of the pump, decreases with increasing filter load. Therefore, change the filter in time, at the latest when displaying the appropriate warning message. A filter load of more than 20 mg should be avoided.

8 Appendix

8.1 Standard protocol

In the 12.50 firmware, the measured particle number concentrations and calculated dust mass fractions are issued in the standard protocol at the same time. The exception is the 1 second interval. Here only particle number concentrations are output, because not all channels are available for calculating the dust mass fractions by halving the measuring range. The output of the measured values with the corresponding preload data (P-line) is done after expiry of an interval.

The twelve dust mass fractions with the line identifier "N" are output at all intervals in $\mu\text{g} / \text{m}^3$ with a resolution of $0.1 \mu\text{g} / \text{m}^3$. The total counts at the end of the N line are output as integer values in the particle / liter unit. The output of the dust mass fractions in the standard protocol occurs allways in the following order and is completed with Total Counts (TC):

TSP, PM₁₀, PM₄, PM_{2.5}, PM₁, PM_{coarse}, einatembar, thorakal, alveolar, pm₁₀, pm_{2.5}, pm₁, TC.

The particle concentrations of the 31 size channels with the line ID "C" are output at intervals of > 6 seconds in the particle / liter unit as integer values. At intervals of 6 seconds or 1 second, the output is in the particle / 100ml unit. The values show cumulative number of concentrations above the respective channel threshold.

An underscore after the line ID indicates the output as an average over the selected measurement interval, e.g. N_ or C_.

A digit (0 to 9) after the line identification indicates the output of the 6-second values, e.g. N0 or C0.

Example: measurement transmission interval:

N_	29.0	19.2	13.6	12.0	7.0	7.2	59.2	20.6	13.8	18.5
11.3	6.3	204769								
C_:	204769	140529	94035	60195	35394	17221	9465	6251		
C_;	2745	1555	1110	675	400	185	100	85		
c_:	39	20	17	13	7	5	5	3		
c_;	1	1	0	0	0	0	0	0		

Example: measurement transmission interval: 6 seconds

N0	29.8	19.7	14.1	12.0	7.0	7.7	57.2	20.1	14.1	18.6
11.4	6.5	214770								
C0:	21477	14053	9404	6020	3540	1722	947	625		
C0;	275	156	111	68	40	19	10	9		
c0:	4	2	2	1	1	1	0	0		
c0;	0	0	0	0	0	0	0	0		

8.1.1 P-Line

There are three data strings in the transmission of measured values. The actual measured values as counts or mass, the P-line with prefix data and the K-line with calibration data.

The K-line appears once upon completion of the self-test at the beginning of each measurement and contains information on the condition of the laser diode and optical measuring cell. A detailed description is contained in chapter 8.1.2.

The P-line is the line marked with a "P" and contains the header data such as date, time, sensor values and some service values. The P-line is output via the interface.

The P-line appears according to the set interval once and looks as follows:

```
P  18   7   17   15   34   33   2   100   0   46   32  26.2  53.5   NA   NA   NA
NA   6  455.9   593.9   51.9 28.1  47.8136  12.9407  464   100.0M   NA
```

The data of the P-line have the following designation:

```
Year Mon Day Hr   Min Sec  Loc GF      Err Qbatt Im Temp rH      Velo Pres Dir
Rain Iv  P_Weight P_Vol[l] rH_i Temp_i Latitude Longitude Height Flow[%] SFlow
```

The meaning of the individual values using sample values (underlined) is explained below.

```
P  18  7  17  15  34  33  2   100   0   46   32  26.2  53.5   NA   NA   NA
NA   6  455.9   593.9   51.9 28.1  47.8136  12.9407  464   100.0M   NA
```

The measurement time with the year, month, day, hour, minute and second appears in the first six places.

```
P  18   7   17   15   34   33  2 100   0   46   32  26.2  53.5   NA   NA   NA
NA   6  455.9   593.9   51.9 28.1  47.8136  12.9407  464   100.0M   NA
```

This is followed by the location number or location selectable from 1 to 99 and the gravimetric factor. For measurements in count mode, the gravimetric factor is not taken into account. In the mass mode, the gravimetric factor is adjustable. The factory setting is 100.

```
P  18   7   17   15   34   33  2   100   0  46   32  26.2  53.5   NA   NA   NA
NA   6  455.9   593.9   51.9 28.1  47.8136  12.9407  464   100.0M   NA
```

Error codes are stored in the ninth place. The error codes are encoded as bit values. The meaning of the error codes is explained in the table under control command E in Chapter 6.3.2. A value of 0 means no error.

```
P  18   7   17   15   34   33  2   100   0   46 32 26.2  53.5   NA   NA   NA
NA   6  455.9   593.9   51.9 28.1  47.8136  12.9407  464   100.0M   NA
```

The state of the power supply is stored in tenth place. This is followed by the motor current consumption of the internal sample air pump. Both values are given in percent. A value of 100 in the power supply indicates that the spectrometer is operated with power supply. Values below 100 reflect the capacity of the battery. Typical values for motor current consumption are between 10 and 40%.

```
P  18   7   17   15   34   33  2   100   0   46   32  26.2 53.5   NA   NA   NA
NA   6  455.9   593.9   51.9 28.1  47.8136  12.9407  464   100.0M   NA
```

The following six values at position 12 to 17 contain the measured values of the optional sensors. These are temperature [°C], relative humidity [%], wind speed [m/s], air pressure [mbar], wind direction [°] and precipitation amount [mm]. If only the analogue sensor is installed, the temperature and relative humidity are stored at the first two places. The remaining values are marked with NA. If the Lufft climate sensor is used, further measurement values are possible depending on the type of model. For further information please contact the nearest Grimm sales office or the appropriate dealer.

The P-line has been adjusted so that the data of the Lufft sensor and the data of the analog sensor are output identically. The output is already converted in °C or %. This eliminates the output of the AD-converted raw data, as well as the output of the analog offsets and factors in the header. To deactivate an

analog channel, its factor and offset must be set to zero. The corresponding channel is then also marked "NA" in the P line.

P	18	7	17	15	34	33	2	100	0	46	32	26.2	53.5	NA	NA	NA
NA	<u>6</u>	455.9		593.9		51.9	28.1	47.8136	12.9407	464		100.0M	NA			

Iv indicates the set storage interval for the measured values.

P	18	7	17	15	34	33	2	100	0	46	32	26.2	53.5	NA	NA	NA
NA	6	<u>455.9</u>		<u>593.9</u>		51.9	28.1	47.8136	12.9407	464		100.0M	NA			

p_weight indicates the calculated dust weight in µg that has passed the measuring cell since the last service / reset, p_vol indicates how many liters of air has passed the measuring cell since the last service / reset.

P	18	7	17	15	34	33	2	100	0	46	32	26.2	53.5	NA	NA	NA
NA	6	455.9		593.9		<u>51.9</u>	<u>28.1</u>	47.8136	12.9407	464		100.0M	NA			

rH_i and temp_i are the calculated relative humidity [%] of the sample air in the measuring cell and the measured temperature [°C] at the measuring cell. These two values are purely informative and indicate the state of the aerosol in the measuring cell.

P	18	7	17	15	34	33	2	100	0	46	32	26.2	53.5	NA	NA	NA
NA	6	455.9		593.9		51.9	28.1	<u>47.8136</u>	<u>12.9407</u>	<u>464</u>		100.0M	NA			

The three values at position 23, 24 and 25, with the connected GPS sensor and receiver, indicate latitude, longitude [decimal degree] and altitude above N.N. [m]. Without GPS signal NA appears.

P	18	7	17	15	34	33	2	100	0	46	32	26.2	53.5	NA	NA	NA
NA	6	455.9		593.9		51.9	28.1	47.8136	12.9407	464		<u>100.0M</u>	<u>NA</u>			

The last two positions (26 and 27) indicate the flow rate (Flow [%]) and the optional flow sensor (SFlow). The letter after the flow indicates the status of the automatic altitude correction ("a" altitude correction activated, "m" altitude correction not activated). The values of the optional flow sensor are specified as the operating volume [l / min]. On the 11-D this is not installed by default. So it always appears NA.

8.1.2 K-line, functional test of the optical measuring cell, the laser diode and photodiode

The self-test verifies the state of the optical measuring cell and the function of the optical components, photodiode in the optical measuring cell and laser diode. The values appear in the so-called K-line. The individual values of the K-line are explained below.

	DC/V	DC_D	DC_H	C0_H	C0_D	LA_L	LA_H
K	665	1292	1654	0	0	59	105

The value DC / v shows the DC voltage of the preamplifier without bias (offset value). This value serves as a reference value and is of no further significance to the customer.

	DC/V	DC_D	DC_H	C0_H	C0_D	LA_L	LA_H
K	665	1292	1654	0	0	59	105

The photodiode generates a "standby voltage" when the laser diode is switched off. This open-circuit voltage is called DC_d, where d stands for "dark". A value of e.g. 1292 in the K-line corresponds to a voltage of 129.2 mV.

When the laser is switched on, the voltage of the photodiode increases, in the example shown above to 165.4 mV. The h at DC_h stands for "high". The increase in DC voltage results from the residual light of the laser, e.g. by reflection in the measuring cell. Most of the laser light is normally absorbed by the light trap.

The DC_d and DC_h values are variable, depending for example on the temperature. The difference between DC_h and DC_d should always remain almost constant - in the above example about 36 mV (165.4 mV - 129.2 mV). Therefore, the difference is a good indicator of the condition of the measuring cell.

If the difference between DC_d and DC_h changes significantly, the optics must be checked and cleaned by the service technician. The nominal difference between DC_d and DC_h can be obtained for each device from the last service protocol or from the acceptance report of the new device.

If the condition of the measuring cell changes, e.g. by contamination, this difference value increases rapidly. A lint trapped in the aerosol inlet nozzle, which extends into the laser beam, may for example increase a DC_h to over 65000. In this case, it is even possible for DC_h to display 0. A difference value close to zero, e.g. of 2.3 mV is typical for a defective laser diode. The defective laser diode emits only weakly, similar to a red LED.

	DC/V	DC_D	DC_H	C0_H	C0_D	LA_I	LA_H
K	665	1292	1654	0	0	59	105

The values C0_h and C0_d must always be 0.

C0_h stands for counts without particles or zero count with the laser switched on, therefore h stands for "high". During the self-test, the measuring cell is flown through by particle-free rinsing air. As a result, no particles can enter the measuring cell and the value must be 0. A non-zero value indicates unwanted particles in the measuring cell, e.g. due to leakage in the sample air flow.

C0_d is detected when the laser is off, so d is for "dark". If there is a value other than zero under C0_d, this indicates an error of the signal amplifier. Since no stray light can occur without laser light, the signal must be caused by electronic noise.

	DC/V	DC_D	DC_H	C0_H	C0_D	LA_I	LA_H
K	665	1292	1654	0	0	59	105

The laser diode in the 11-D is operated alternately with two laser outputs, the so-called multiplex mode. La_l "Low" means low laser current in MA and La_h "high" high laser current. If the La_h current increases to a value greater than 180 MA, the laser diode is probably defective.

For all devices a Quality Assurance Record is provided, which for comparison purposes contains the values of the K-Line at the factory calibration by the manufacturer.

8.2 RS-232 Commands

A	(Alarm) Output of the current alarm threshold. In standby this can be changed.																																
B	(Battery) Output of battery capacity. When the power supply is connected, the value is always 100%.																																
E	(ERROR) Output error code <table><tr><th>Error-Code</th><th>Display output</th><th>Meaning</th></tr><tr><td>128</td><td>NEW SELFTEST</td><td>Self-test failed</td></tr><tr><td>64</td><td></td><td>Not used</td></tr><tr><td>32</td><td></td><td>Not used</td></tr><tr><td>16</td><td>BATTERY TOO LOW OUT OF WORK</td><td>Battery 0 %</td></tr><tr><td>8</td><td>BATTERY LOW PLEASE RECHARGE</td><td>Battery < 10 %</td></tr><tr><td>4</td><td></td><td>Not used</td></tr><tr><td>3</td><td>CHECK DUST FILTER AND AIR PASSAGE</td><td>Sample volume flow control outside the control range</td></tr><tr><td>2</td><td>CHECK DUST FILTER AND AIR PASSAGE</td><td>Pump current < 15 %</td></tr><tr><td>1</td><td>CHECK DUST FILTER AND AIR PASSAGE</td><td>Pump current > 70 %</td></tr></table>			Error-Code	Display output	Meaning	128	NEW SELFTEST	Self-test failed	64		Not used	32		Not used	16	BATTERY TOO LOW OUT OF WORK	Battery 0 %	8	BATTERY LOW PLEASE RECHARGE	Battery < 10 %	4		Not used	3	CHECK DUST FILTER AND AIR PASSAGE	Sample volume flow control outside the control range	2	CHECK DUST FILTER AND AIR PASSAGE	Pump current < 15 %	1	CHECK DUST FILTER AND AIR PASSAGE	Pump current > 70 %
Error-Code	Display output	Meaning																															
128	NEW SELFTEST	Self-test failed																															
64		Not used																															
32		Not used																															
16	BATTERY TOO LOW OUT OF WORK	Battery 0 %																															
8	BATTERY LOW PLEASE RECHARGE	Battery < 10 %																															
4		Not used																															
3	CHECK DUST FILTER AND AIR PASSAGE	Sample volume flow control outside the control range																															
2	CHECK DUST FILTER AND AIR PASSAGE	Pump current < 15 %																															
1	CHECK DUST FILTER AND AIR PASSAGE	Pump current > 70 %																															
G	(Gravimetry C-factor) Output of the gravimetric factor. In standby mode the C-factor can be changed between 0.1 and 9.8 in steps of 0.1. The default setting is beträgt 1,0. The C-factor is taken into account for the calculation of the following six mass fractions: inhalable, thoracic respirable, pm ₁₀ , pm _{2.5} , pm ₁ .																																
H	(Hour) Output of operating hours																																
I	Interval Output of the storage interval for storing the measured values in the data memory. Changing the memory interval in standby: 0 = 1 Minute 1 = 5 Minutes 2 = 10 Minutes 3 = 15 Minutes 4 = 30 Minutes 5 = 60 Minutes 6 = 6 Seconds 7 = 1 Second When changing to interval 7 (1 second), the high laser power is automatically selected. A change to the low laser line for the upper 16 channels is possible via ">".																																
J	(Output Channels) Output format of the measured values																																
L	(Location Number) Output of the location number and change in standby.																																
M	(Mean) Output means and related sample volume.																																

R	(Run) Start the measurements from standby.
S	(Standby) Completion of the measurement; Change to standby.
T	(Time) Output of the device time and set the clock in standby mode. Changing the minutes sets the seconds to zero.
U	(Unlock) Keyboard lock U=0 Keyboard free U=1 Standby cannot be taken U=2 No keyboard operation possible
V	(Version) Output the version number of the firmware
W	(Weight) Output calculated filter weight and related sample volume
Z	(Zero Clear Mean) Output and subsequent resetting of the mean values and the volume
@	(Serial-No.) Output of the serial number
Long Break	(Power ON) Turn on the instrument (if off with ^ Y)
—	(Output User Strings + Factors) Output of user texts and calibration factors of the analog values of the sensor. The analog voltage signals are converted into the measured variable with a factor for the slope and a factor for the offset value.
*	(User Factors (Analog Inputs)) Change user factors and offset (only in standby).
!	(Output Model + Rev + Version) Output of model, firmware revision and version.
;	(Autocal) Enter automatic restart with self-test every X hours. For X, values from 1 to 99 hours can be set. 0 turns off the function.
\	(IP address) Request or change the IP address. Information on changing the IP address can be found in chapter 6.3.2. TIP: If the IP address is unknown, it can be first queried / changed via a HyperTerminal connection via a COM port.
' or "	(Commentary until CR) A comment can be entered. The function is ended by ENTER.
\$	(User Strings (Analog Input)) Selection or modification of the user texts as well as the factors for the analog voltage outputs on the display (in standby mode only). Special characters (via ASCII 127) cannot be entered. During output, the '"' character (ASCII 248) is output as ' _ '. The tabulator key can be used to select predefined texts that are confirmed by the enter key. They can also be overwritten with any text. The number output is always five digits and starts from the ninth digit. If characters are entered here, except for a decimal point, an acoustic warning is issued. The multiplication factor related to 1 V can then be entered after the text. It can also be changed by the command `*`. Example: Temperature sensor: 0 °C = 3.0V and 50 °C: =: 8.0V User-Text: Temp .: °C

	User factor: 10.0 [° C / V] Offset: 3.00V The sensor data are read out directly from the climate sensor (the sensors starting with the identifier "E", as the 3rd character in the serial number, are equipped with an EEPROM for the sensor data). The transfer of the data from the EEPROM, including the user text, takes place only if the user factors in the device are set to 1.0 and the offset values 0.0. Recalibration by the user is therefore still possible (positive offset values are not possible).
?	(Help) Output of help with all commands <pre>##### Help for Dust Monitor ##### A Alarm B Battery E Error G Gravimetry C-Factor H Runtime hours I Interval J Output Channels T Time Set [Standby] ^T Timer Set [Standby] L Location Code [1...99, +] ^L Land (for Date) [Standby] M Mean Value Z Zero Clear Mean R Run Measurement S Standby Modus U Unlock/Lock Keys [Standby] @ Serial-No. V Version W Weight \$ User Strings (Analog Inputs) * User Factors (Analog Inputs) _ Output User Strings + Factors ! Output Model + Rev + Version \ IP-Address ; Autocal : Gesytec [Standby] ' Commentary until CR " Commentary until CR ^D Disable Output ^E Enable Output; no Fast ^Y Power OFF Long Break: Power ON C Count Mode [Standby] < only Count_Channel 1...16 > only Count_Channel 16...32 #####</pre>

All the following commands are made with {Ctrl} + {Letter}

^D	(Disable Output) Disables the measured value transmission.
^E	(Enable Output, no Fast) Switching on / off the fast mode transmission of the measured value
^L	(Land (for Date)) Country setting (E or U) for date output on the instrument (standby only). 0 EU 1 US
^T	(Timer Set) Timer operation for automatic switching on and off of the dust measuring device
^Y	(Power OFF) Switch off the device

Service mode

The 11-D can be switched into different service modes. Service mode 0 is a service mode that is also accessible to customers. All higher quality service modes are only available to service technicians or after appropriate training. The following refers only to the service mode 0.

The service mode offers the following functions or differences:

- In service mode additional control commands are available, e.g. the last service query with "~" last service.
- Additional information is displayed for selected control commands.
- When outputting data via the RS-232 interface, additional data and explanatory texts are output via the P-line.

Switching to the service mode can only be done in standby mode, not in the current measuring mode.

Switching on the service mode

Pressing the pipe symbol "|" on the> <key with {AltGr} {> <} and then pressing the {Tab} key switches the instrument into the service mode.

Switching off the service mode

By pressing the pipe symbol "|" on the> <key with {AltGr} {> <} and then pressing any key, the service mode is ended.

Other commands in service mode

^B	(Baud rate) Setting the data transfer rate 0 = 9.600 1 = 19.200 2 = 57.600
^F	(Flow Adjust) Activation of automatic height correction can be checked with the command ^ F in service mode. Flow: 100.0 Im: 59.8% m P: 1008 hPa Flow: 100.0 Im: 59.8% a P: 1008 hPa The letter after the flow indicates the status of the automatic altitude correction ("a" altitude correction activated, "m" altitude correction not activated). If altitude correction is not active, the flow can be adjusted in steps of 0.05 l / min with {+} and {-}. At the end of the line, the absolute pressure is given in hPa.
~	(Last Service) Last Service. The last service – by whom and when performed. Identification by individual service key and date.
&	(Mixer Humidity Threshold) This command is not active on 11-D
[	(Position & GPS) With the GPS sensor and GPS receiver, longitude, latitude in decimal degrees and altitude in m are output via N.N.
]	(Laser_Current_Trend) The laser current is stored in the EEPROM every 1000 operating hours. The] command outputs these stored values.
?	(Help) Output of help with all commands in service mode 0 (Help) <pre> ##### Help for Dust Monitor ##### A Alarm B Battery E Error G Gravimetry C-Factor H Runtime hours I Interval J Output Channels T Time Set [Standby] ^T Timer Set [Standby] L Location Code [1...99, +] ^L Land (for Date) [Standby] M Mean Value Z Zero Clear Mean R Run Measurement S Standby Modus U Unlock/Lock Keys [Standby] @ Serial-No. V Version W Weight \$ User Strings (Analog Inputs) * User Factors (Analog Inputs) _ Output User Strings + Factors ! Output Model + Rev + Version \ IP-Address ; Autocal : Gesytec [Standby] ' Commentary until CR " Commentary until CR ^D Disable Output ^E Enable Output; no Fast ^Y Power OFF Long Break: Power ON C Count Mode [Standby] < only Count_Channel 1...16 > only Count_Channel 16...32 ~ last Service X X-tal ^Z P_Weight=0, P_Volume=0 ^F Flow Adjust [Standby] [Position & GPS [Standby] & Mixer Humidity Threshold ^B Baudrate L Location Code = 0 : Run_Nr.= 0] Laser_Current_Trend ##### </pre>

8.3 Calculation of the dust mass fractions

The predecessor models of the 11-D and other mobile or stationary Grimm aerosol spectrometers supplied dust mass fractions with the same name, but based on different calculations. This sometimes led to difficulties in comparing readings.

The following table explains the calculations of the dust mass fractions of the 11-D.

Table 13: Parameters for calculation of the dust mass fraction in the 11-D

value	[1] extra- polated channel	[2] spherical shape	[3] variable density	[4] uniform density	[5] C- factor	[6] SVC- correction	[7] volume flow correction	[8] EN 481 conformity
TSP	✓	✓	✓	never	never	✓ ¹	✓ ¹	never
PM ₁₀	✓	✓	✓	never	never	✓ ¹	✓ ¹	never
PM ₄	✓	✓	✓	never	never	✓ ¹	✓ ¹	never
PM _{2.5}	✓	✓	✓	never	never	✓ ¹	✓ ¹	never
PM ₁	✓	✓	✓	never	never	✓ ¹	✓ ¹	never
PM _{coarse}	✓	✓	✓	never	never	✓ ¹	✓ ¹	never
Inhalable	✓	✓	never	✓	✓	never	never	✓ ²
Thoracic	✓	✓	never	✓	✓	never	never	✓ ²
Respirable	✓	✓	never	✓	✓	never	never	✓ ²
pm ₁₀	✓	✓	never	✓	✓	never	never	never
pm _{2.5}	✓	✓	never	✓	✓	never	never	never
pm ₁	✓	✓	never	✓	✓	never	never	never
Consistant with	all GRIMM 11x and EDMx	all GRIMM 11x and EDMx	107 11-E EDM164 EDM180	1108 1109 11-A 11-C 11-R	1108 1109 11-A 11-C 11-R	107 11-E EDM164 EDM180	107 11-E EDM164 EDM180	1108 1109 11-A 11-C 11-R

¹: The correction is active only, if the analog sensor for T and RH is connected and therefore provides the required temperature information.

²: Conformity is given only, if the radial symmetric sampling inlet is used or the required inlet velocity of 1.25 m/s is ensured.

Comments

[1] Extrapolated channel: one size channel is extrapolated below the detection limit (0.25 µm). The result of this extrapolated channel is used for the calculation of all dust mass fractions.

[2] Spherical shape: All mass calculations based on the assumption of spherical particles to calculate the particle volume. For each size channel the arithmetic mean particle diameter is used.

[3] Variable density: The Enviro PM values are calculated with variable particle densities. The densities are different for the size channels. They depend on the particle size and are based on typical aerosol composition for defined particle sizes and gravimetric comparative measurements in the ambient air. Note: These values will not be published or shared.

[4] Uniform density: The occupational medical values are calculated with a uniform density. This is identical for all size channels. It is based on gravimetric comparison measurements of a test aerosol. Note: This value will not be published or shared.

[5] C-factor (gravimetric factor): For a known aerosol or under defined conditions, a correction factor can be determined by weighing the internal PTFE filter as follows:

$$\begin{aligned} \text{C-factor} &= \text{actual dust mass on the filter} / \text{dust mass calculated by the 11-D} \\ &= (\text{empty filter} - \text{loaded filter}) / \text{P-Weight} \end{aligned}$$

Via HyperTerminal, 1179 spectrometer software or Modbus the C-factor can be changed.

[6] SVC Correction: This reduces the original dust mass by an amount resulting from the loss of semi-volatile components (long sampling, uncooled filters) during sampling and uncooled transport of the filters on gravimetric dust collectors.

The SVC correction is only performed if an analog sensor for T and RH is connected and supplies the temperature value necessary for the calculation.

Note: for models 107, 11-E (EDM164 or EDM180), it is not possible to measure without analogue sensor for T and RH. The use of a simulator plug provides values that are different from the analog sensor values for T and RH.

[7] Volume flow correction: This normalizes the sample volume to the relationship between the outside temperature and the device temperature (20° C), which is typical for permanently installed devices in containers. The volume flow correction only takes place if an analogue sensor for T and RH is connected and supplies the temperature value necessary for the calculation.

Note: for models 107, 11-E (EDM164 or EDM180), it is not possible to measure without analogue sensor for T and RH. The use of a simulator plug provides values that are different from the analog sensor values for T and RH.

[8] EN 481 compliant: This standard defines a suction rate for the measurement of occupational medical dust mass fractions of 1.25 m/s. Conformity with EN 481 is only fulfilled when measured with the radial symmetrical sampling head or a defined suction velocity of 1.25 m/s.

9 Index

accessories	15	LAN.....	18, 26
acclimatization time	9	laser	9, 14, 37
alarm	20	measuring mode.....	21
alarm threshold	20, 40	Modbus.....	30
analog inputs	14	name plate.....	24
analog offset	14, 41	operation	14, 19
analog slope	14, 41	P-line	36, 37
battery	23	power supply	14, 23
Bluetooth.....	10, 24, 26	PTFE-filter	12, 35
country setting	43	reproducibility	14
damage.....	9	rinsing air	6
data output.....	14	RS-232	25
data storage.....	14	RS-232 commands.....	33, 40
Datenausgabe	36	RS-232 Pin assignment	25
diluter	16	sampling head	22
display.....	14, 18	self-test.....	6, 37
disposal.....	12	service mode	43
Ethernet	6, 18	size channels.....	14
firmware	41	spectrometer software 1179.....	28
GPS	24, 38, 44	standby.....	21, 41
height correction	44	status LED	23
HyperTerminal	32	Staubmassenfraktionen	45
interval	21, 22	storage interval.....	14
IP address changing.....	33	switch off	43
IP-address	30	USB	25
key functions.....	19, 21	USB flash drive.....	15, 24
keyboard lock	41	volume flow rate	14
K-line.....	37	warranty.....	24